What Laya is: a reverse-engineering teardown of the open decision model that Themis must beat

PinoAI engineering note · THEMIS-ENG-001 · technical report · v1.2 (erratum: section 4a interface keys)

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2026-10-07

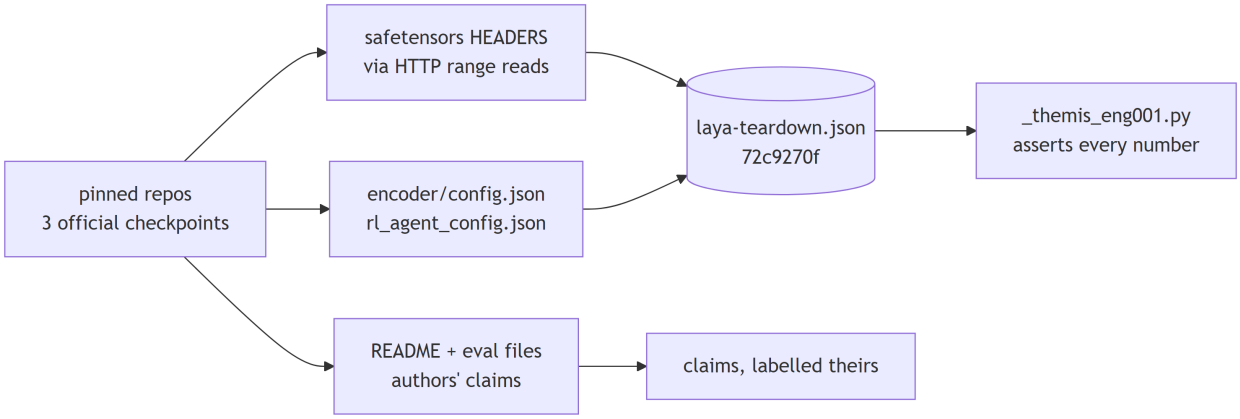
Abstract

Themis, PinoAI's in-house judge, is designed to replace a commercial decision service and to beat Laya, the open (Apache-2.0) model built for the same job. Before designing Themis we took Laya apart: three official checkpoints at pinned revisions, read from their configuration files and safetensors headers without downloading a weight, plus Laya's own code, README and evaluation files.

1. **It is an encoder with a small decision head, not a language model.** English: ModernBERT-large, 28 layers, hidden 1,024, vocabulary 50,368, 421,293,830 parameters. Multilingual: mmBERT-base, 22 layers, hidden 768, vocabulary 256,000, positions to 8,192, 321,908,998 parameters. The head has 2 layers and scores each option at its own mask token in one forward pass. The typed-decisions checkpoint ships all-F16; the head runs under bf16 autocast.
2. **Abstention is part of the interface, with costs.** The head can ESCALATE instead of answering; the config prices an escalation at 0.5 and a wrong action at 3.0. Any comparison must therefore score coverage, not only accuracy.
3. **A shared option budget is a structural limit by design.** Every option in a question shares `head_max_len` (192 tokens on English, 256 on multilingual); at 77 options that is (256 − 16) // 77 = 3 tokens per option. Laya's authors report 0.425 on Banking77 against the commercial judge's published 0.870. What actually drives Laya's many-option collapse is measured in THEMIS-BENCH-001: on the same 100 items, the option wording/label-set change dominates (72 BTZSC sentence-style options 0.02 vs 77 short PolyAI category names 0.37; paired McNemar 37/2 discordant, p = 2.8e-09), while a 4x larger budget adds a small, significant gain on the multilingual route (0.02 -> 0.09, 7/0 discordant, p = 0.016). A "stripped prefix" row turned out ineffective (the strip removed one word; the shared-prefix hypothesis is untested - the 98/100 collapse onto option 0 confounds position with opening salience, BENCH-001 v1.1). The budget is kept here as arithmetic: a real limit, not the main driver.
4. **Routing is by script.** Non-Latin text goes to the multilingual checkpoint; Latin-script text stays on the English checkpoint unless a language guess is supplied. Tagalog and Taglish are Latin-script.
5. **Laya's own "Honest Limits"** (its README) add: base checkpoints near chance zero-shot on typed decisions (the 0.766 belongs to a checkpoint fine-tuned on that benchmark's own training split); ordinal scoring the weakest primitive; a yes/no head that can follow its option labels; an abstention probability with "no usable signal yet"; overconfidence that needs per-family temperature fitting; and an 11+-option temperature the loader clamps and declares uncalibrated.

Every item is a Themis design target (THEMIS-RE-DESIGN); the registered gate remains superiority over Laya as released at its best configuration, plus agreement with human labels.

1. Method



Static teardown: pinned repositories, headers and config only; probes run separately (THEMIS-BENCH-001).

`scripts/judge/laya_teardown.py` resolves each registered revision, reads the licence from the card at that pin, lists the files, parses the encoder and head configs, and counts parameters from the safetensors headers by HTTP range reads. No weight was downloaded and nothing was executed from the repositories. The third-party fine-tune `laya-grounded` (CC-BY-NC-4.0) is excluded. Behavioural probes (harness match, option framing, routing) are a separate note so that this one stays a statement of what the artefact is.

2. The three checkpoints

Table 1: The official checkpoints at their registered pins.

Checkpoint	Revision	Licence at pin	Encoder	Params	Head budget
`convaiinnovations/laya` (English)	55cf4c4e	Apache-2.0, not gated	ModernBERT-large: 28 layers, hidden 1,024, vocab 50,368	421,293,830 (the root repo also bundles the other two: 1,164,496,658 across three files)	max_len 512, head_max_len 192
`laya-multilingual`	e4e9ddf2	Apache-2.0, not gated	mmBERT-base: 22 layers, hidden 768, vocab 256,000, positions 8,192	321,908,998	max_len 1,024, head_max_len 256
`laya-typed-decisions`	1a793eb5	Apache-2.0, not gated	ModernBERT-large (fine-tuned from the English checkpoint)	421,293,830, all F16	max_len 1,024, head_max_len 256

Shared head configuration (all three): 2 head layers, up to 6 prefixes, act costs {escalate: 0.5}, wrong-act cost 3.0, autocast dtype bf16.

3. How a decision is made

The state (text, email, ticket or JSON) and the typed question are packed into one sequence: the state within `max_len − head_max_len`, the options within `head_max_len`. Each option is rendered to its own mask token; the head scores the masks and a softmax over that question's options gives calibrated probabilities after per-family temperatures. An act/escalate head decides whether to answer at all. Everything is one forward pass, which is why Laya is fast (its authors report ~33 ms per call). The design choices that matter for Themis:

- **The option budget is shared.** It is a structural limit by design (3 tokens per option at 77 options), but it is not the main measured driver of the many-option collapse: THEMIS-BENCH-001 found the wording/label-set change dominates (paired p = 2.8e-09) and a 4x budget adds a small significant gain (+7 points, p = 0.016, multilingual route only). Themis scores options with full context (option-wise or hierarchical) so that neither the option count nor the option wording can erase the distinctions between options; that target is motivated by the wording result.
- **Abstention has a price but, by Laya's own account, no usable signal.** Themis trains its abstention head on correctness and is compared at matched coverage.
- **Calibration is a post-step the user must run.** Themis ships fitted temperatures.

- **Script routing.** Themis uses one multilingual model or real language identification, so Filipino never meets an English-only vocabulary by default.
- **Silent precision fallback.** On GPUs below compute capability 8 the head runs in fp16; our Themis arms pin their dtype and record it, and any Laya comparison runs Laya in bf16 or fp32.

4. What Laya's authors claim, and what we take from it

Laya's README compares it with the commercial judge using third-party published numbers (never measured by the authors, who had no API access): typed-decisions 0.766 vs 0.727; AG News 0.950 vs 0.910; DAIR Emotion 0.595 vs 0.480; Banking77 0.425 vs 0.870; ECE 0.081 vs 0.246. We reproduce none of the commercial judge's numbers ourselves and cite them only as third-party published, unverified, not paired; Laya's own numbers we test in THEMIS-BENCH-001 before relying on them.

4a. The interface Themis targets

Themis's contract is the REGISTERED typed-decisions schema (PH-JUDGE-V1-PLAN: `choice` / `score` / `noul` plus the escalate action with calibrated probabilities), not either of Laya's two output surfaces. Laya exposes (i) an in-process answer object and (ii) the `laya-serve` HTTP response on `POST /v1/systemone`. Read live from laya 0.3.28 (in-process keys: ml-engineer's two live reads, 3a46e7 and a37ea8; HTTP keys: devops's REA pilot observation, a186fe), BOTH surfaces carry the same keys: `action` (with `act_probability`), `answer_confidence`, `choice`, `confidence`, `probabilities` and `type`, plus `usage.truncated` in process. **Erratum (v1.2):** v1.1 said the in-process surface was the `rl_agent` / `system_one` shape with a top-level `act_probability`, distinct from the HTTP keys; that came from the README, not from running the code, and is wrong for 0.3.28. Themis ships one documented adapter for this shared shape, and any "drop-in" claim names the Laya version it matches. The HTTP shape is the one existing clients of the commercial judge use, so it is the adapter that matters for replacement.

5. Limitations

- Static read: parameter counts and configs are exact; behaviour (routing, framing sensitivity, precision effects) is established separately by probes.
- Laya's README claims are the authors'; sample sizes and prompts differ from ours.
- The commercial judge's internals are unknown by design and stay so.

Reproducibility

Table 2: Files.

Item	Where
Input	`experiments/results/themis/laya-teardown.json` (sha256 72c9270f…; git-ignored result, rendered in the Lab room /themis)
Generator	`scripts/judge/laya_teardown.py` (sha256 fdf51b64…, registered)
Numbers in this note	`experiments/reports/_themis_eng001.py` prints and asserts each one
Narrative companion	`experiments/reports/LAYA-TEARDOWN-note.md` (registered)
Design response	`experiments/plans/THEMIS-RE-DESIGN.md`; registered plan `data/eval/experiments/PH-JUDGE-V1-PLAN.md`

Acknowledgments

We thank the leadership of Kooya PH for supporting this work, and the native-speaker evaluators who reviewed and rated the Filipino and Taglish material.

AI assistance. Analysis code, statistical checks and drafts of this document were prepared with the help of Claude (Anthropic), an AI model. Every number is generated from the project's result files. The author reviewed the content and takes responsibility for it.